

*One argument, not seven units: choose a loss, minimise it over a class of functions, on columns you chose, and then prove it on data the fit never saw*

## 1. What are we minimising?

a loss  $L(y, \hat{y})$

MSE fits the mean  
MAE fits the median  
cross-entropy fits  
a probability

*Unit II*

## 2. How is it minimised?

step downhill

$$w \leftarrow w - \eta g$$

step size, batch size,  
momentum, per-coordinate  
rates

*Unit III*

## 3. What is being fitted?

a hypothesis class

line, sigmoid, tree,  
forest, network, margin

same machinery,  
different shape

*Units V and VI*

## 4. How would I know?

data the fit never saw

$k$ -fold, nested CV,  
confusion matrix, ROC

a score is an estimate  
with a spread

*Unit IV, and Unit VI*

## 0. What do you feed it?

clean, impute, scale, encode, engineer, select

every one of these learns a number from the data,  
so every one of them belongs inside the fold

*Unit IV*

## And with no target column?

delete  $y$  and boxes 1 and 4 both vanish:  
there is nothing to score against.  
distance replaces the loss, and the burden  
of stating your assumptions is entirely yours

*Unit VII*